

Experiment 6. Case study on Input data analysis

Aim: To create and analyze a **histogram plot in MATLAB** using numerical and categorical data, and to study the distribution of given data values.

Procedure:

1. Start MATLAB

- Open MATLAB and go to the Command Window or create a new script file.

2. Define Data

- Enter the dataset as a vector (as shown in the image):

```
tvData = [60, 90, 30, 0, 15, 13, 97,  
45, 25];
```

3. Plot Basic Histogram

- Use the histogram() function to visualize data distribution:

```
histogram(tvData)
```

4. Modify Number of Bins

- Change the number of bins:

```
histogram(tvData, 3)
```

5. Specify Bin Ranges

- Define custom bin intervals:

```
histogram(tvData, [0, 30, 60, 120])
```

6. Categorical Data Histogram

- Convert numeric data into categories:

```
A = [0 0 1 1 1 0 0 0 NaN NaN 1 0 0  
0 1 0 1 0 1 0 0];
```

```
C = categorical(A, [1 0 NaN],  
{'yes','no','undecided'});
```

```
histogram(C)
```

7. Run the Code

- Execute the commands and observe histogram outputs in the figure window.

Code:

```
% Numerical Data Histogram
```

```
tvData = [60, 90, 30, 0, 15, 13, 97, 45, 25];
```

```
% Basic histogram
```

```
figure;
```

```
histogram(tvData);
```

```
title('Basic Histogram of tvData');
```

```
% Histogram with 3 bins
```

```
figure;
```

```
histogram(tvData, 3);
```

```
title('Histogram with 3 Bins');
```

```
% Histogram with custom bin edges
```

```
figure;
```

```
histogram(tvData, [0, 30, 60, 120]);
```

```
title('Histogram with Custom Bins');
```

```
% Categorical Data Histogram
```

```
A = [0 0 1 1 1 0 0 0 NaN NaN 1 0 0 0 1 0 1 0  
1 0 0];
```

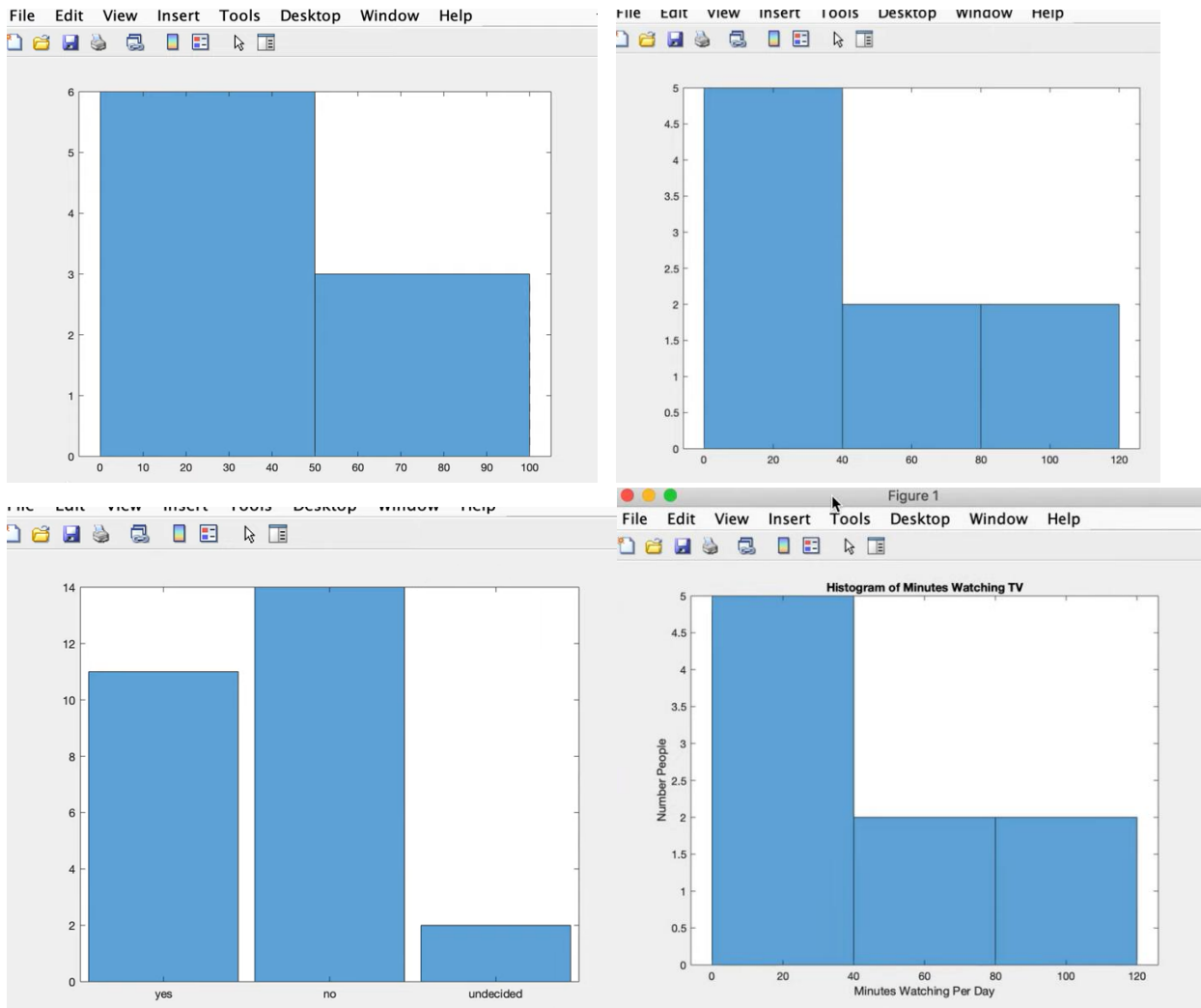
```
C = categorical(A, [1 0 NaN],  
{ 'yes','no','undecided' });
```

```
figure;
```

```
histogram(C);
```

```
title('Categorical Histogram');
```

Results:



Conclusion:

The experiment demonstrates how MATLAB's `histogram()` function is used to visualize data distribution.